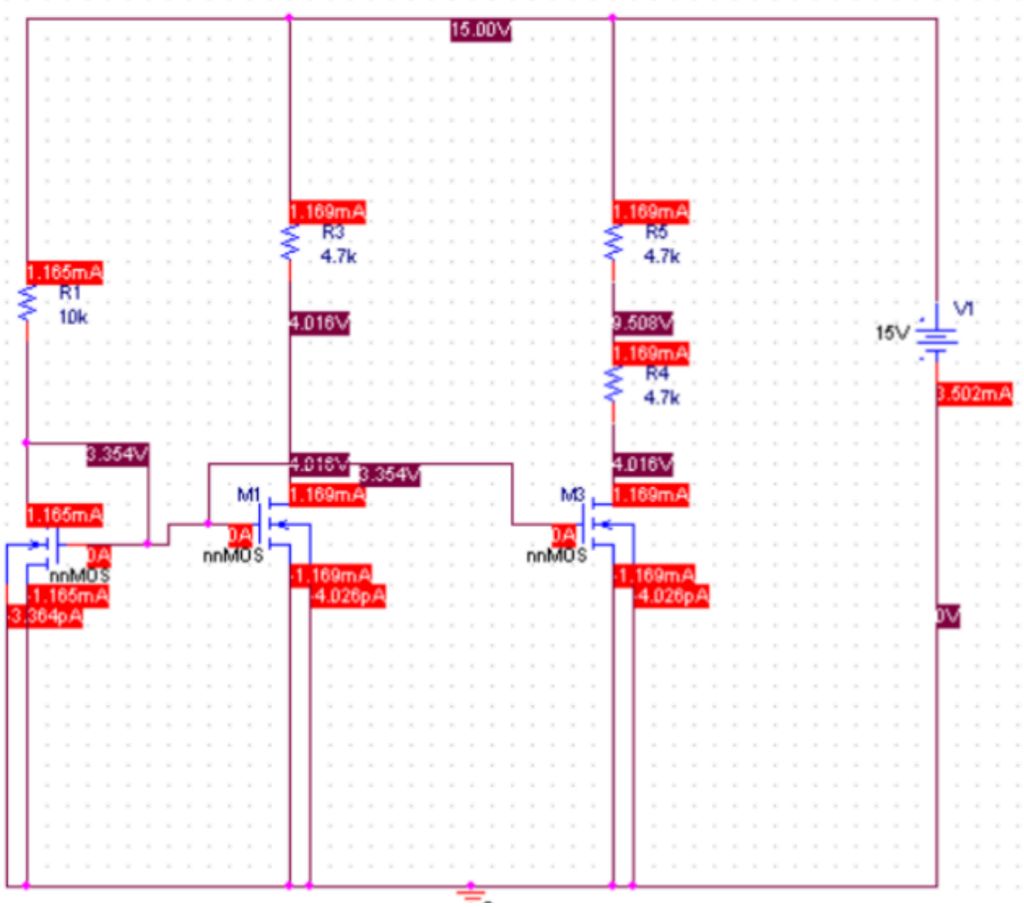
Thomas Puskas

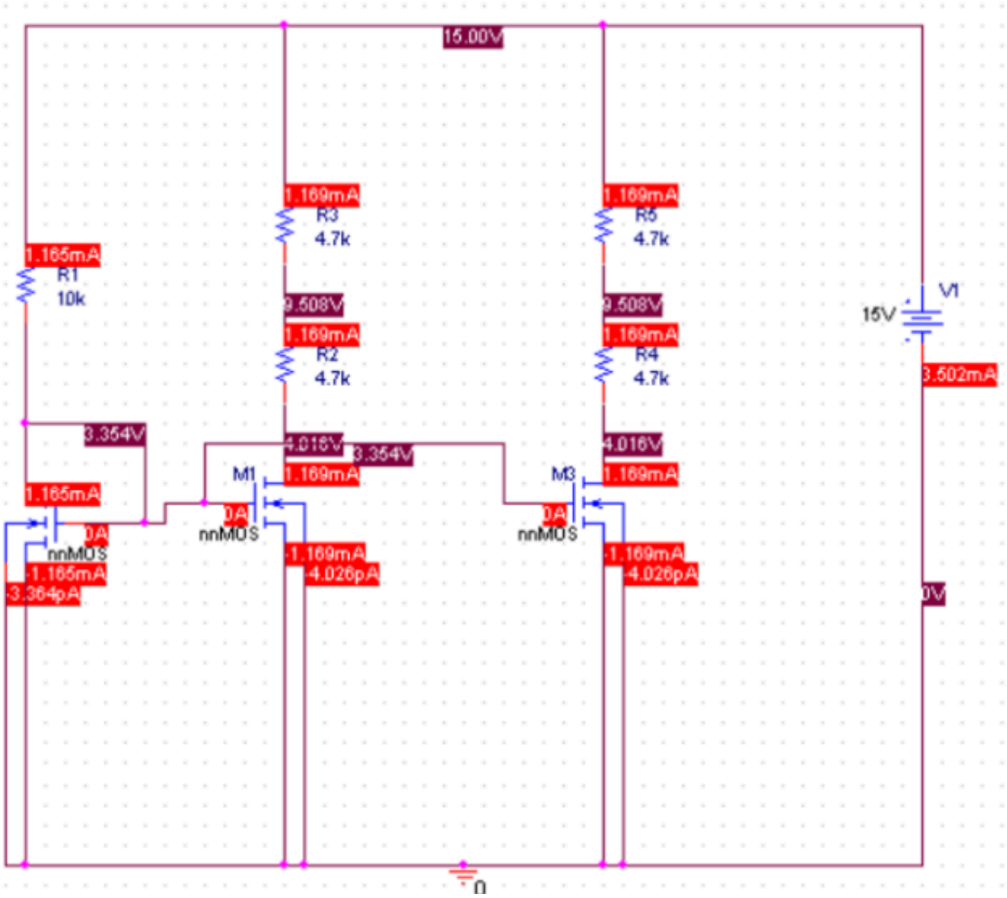
2025952293

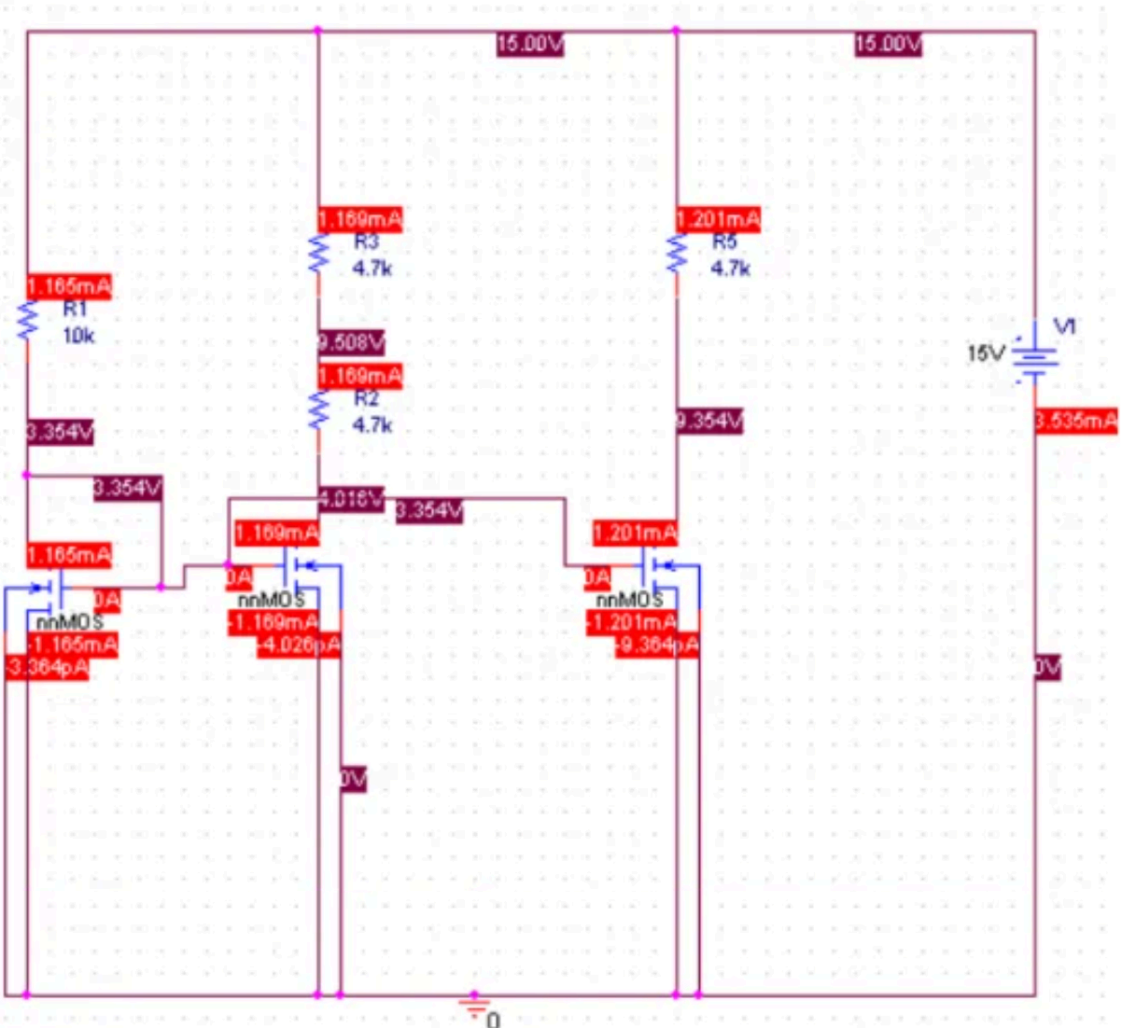
The purpose of this experiment is to understand and look at current mirror circuits which based on a current voltage and curruent source are able to offer a bias. We also observed the advantages and disadvantages of the basic current mirror circuit and the cascode current mirror circuit and we observed the advantages of the Wilson current mirror which adops the negative current feedback.

Pre lab:

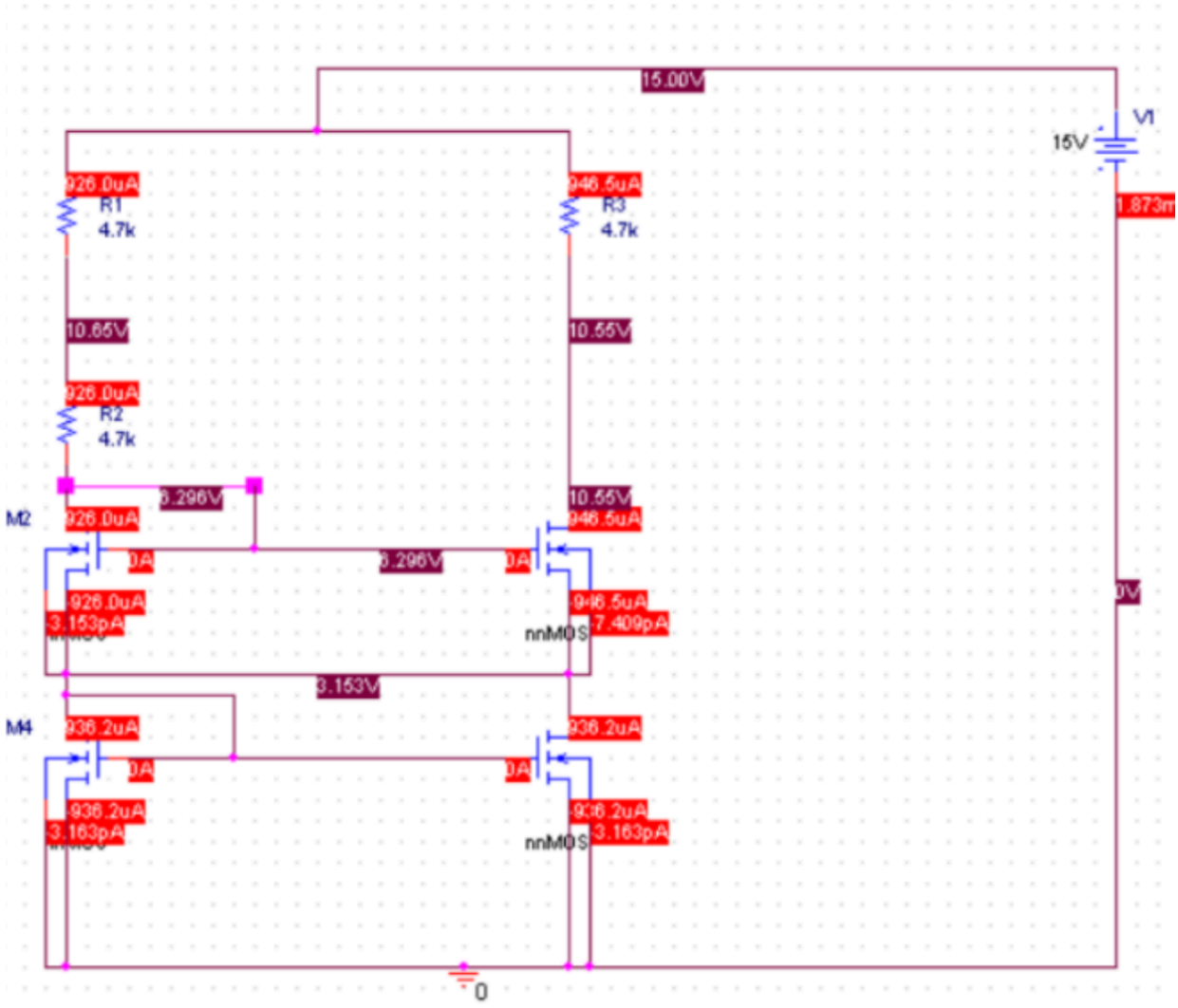
Experiment 11.1

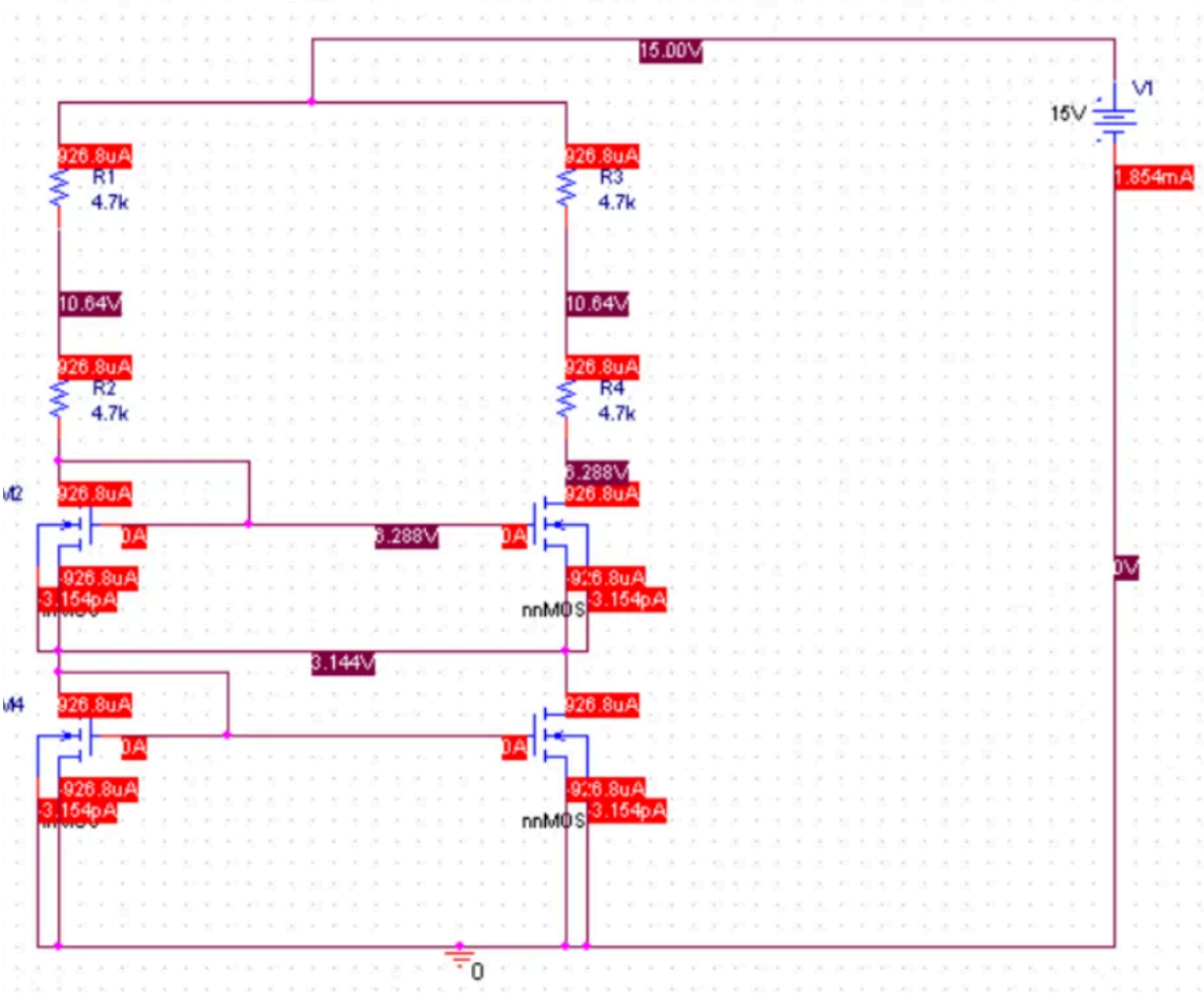


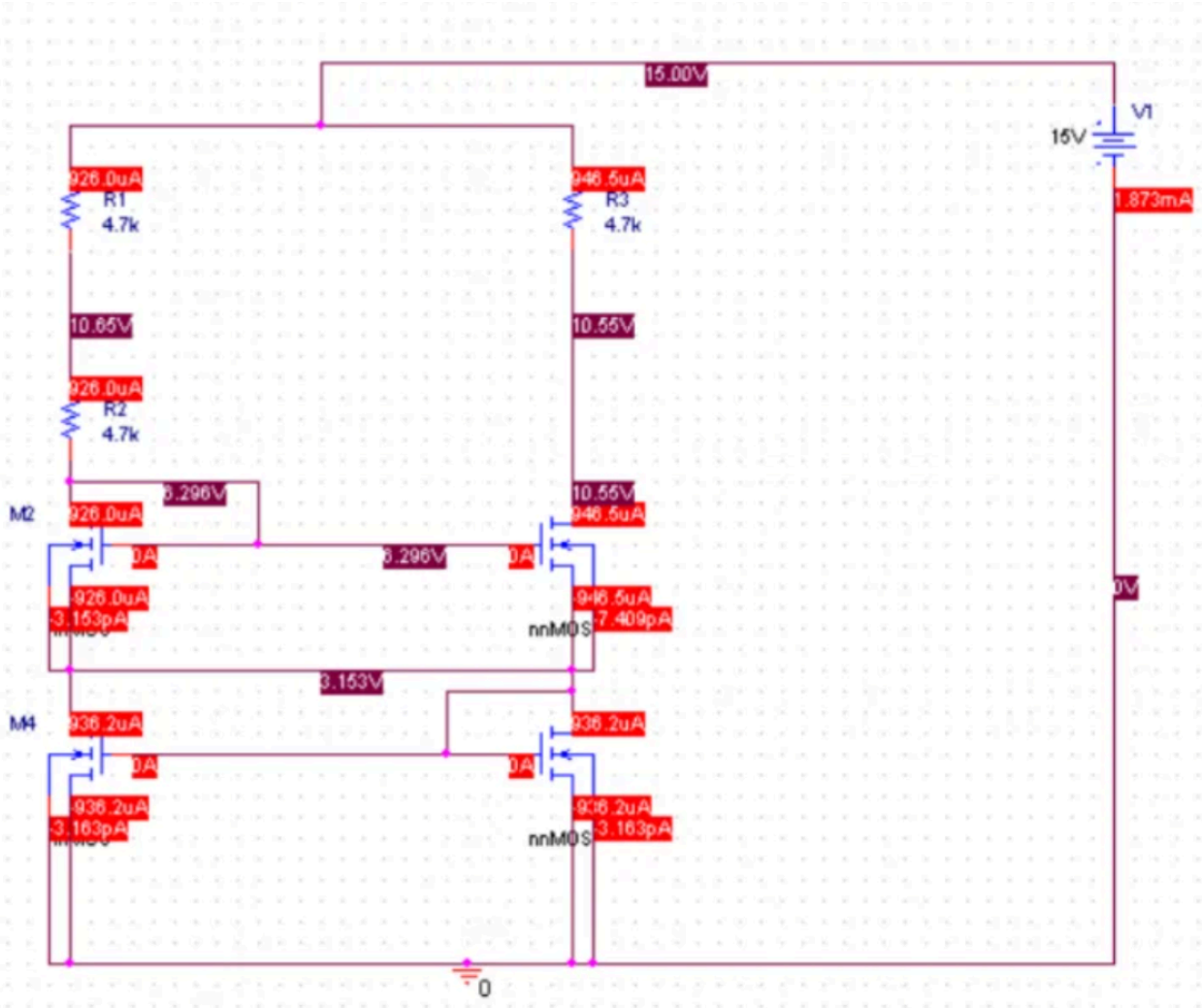

15.00V
1.169mA
R3
4.7k
1.169mA
R5
4.7k
1.165mA
R1
10k
9.508V
1.169mA
R2
4.7k
9.508V
1.169mA
R4
4.7k
V1
15V
3.502mA
8.354V
4.016V
8.354V
M1
1.169mA
4.016V
M3
1.169mA
1.165mA
nnMOS
nnMOS
1.169mA
4.026pA
1.165mA
3.364pA
1.169mA
4.026pA
0V
0


15.00V
15.00V
1.169mA
R3
4.7k
1.201mA
R5
4.7k
1.165mA
R1
10k
9.508V
1.169mA
R2
4.7k
V1
15V
3.354V
9.354V
3.354V
1.165mA
1.169mA
1.201mA
nnMOS
nnMOS
nnMOS
-1.165mA
-3.364pA
-1.169mA
-4.026pA
-1.201mA
-9.364pA
0

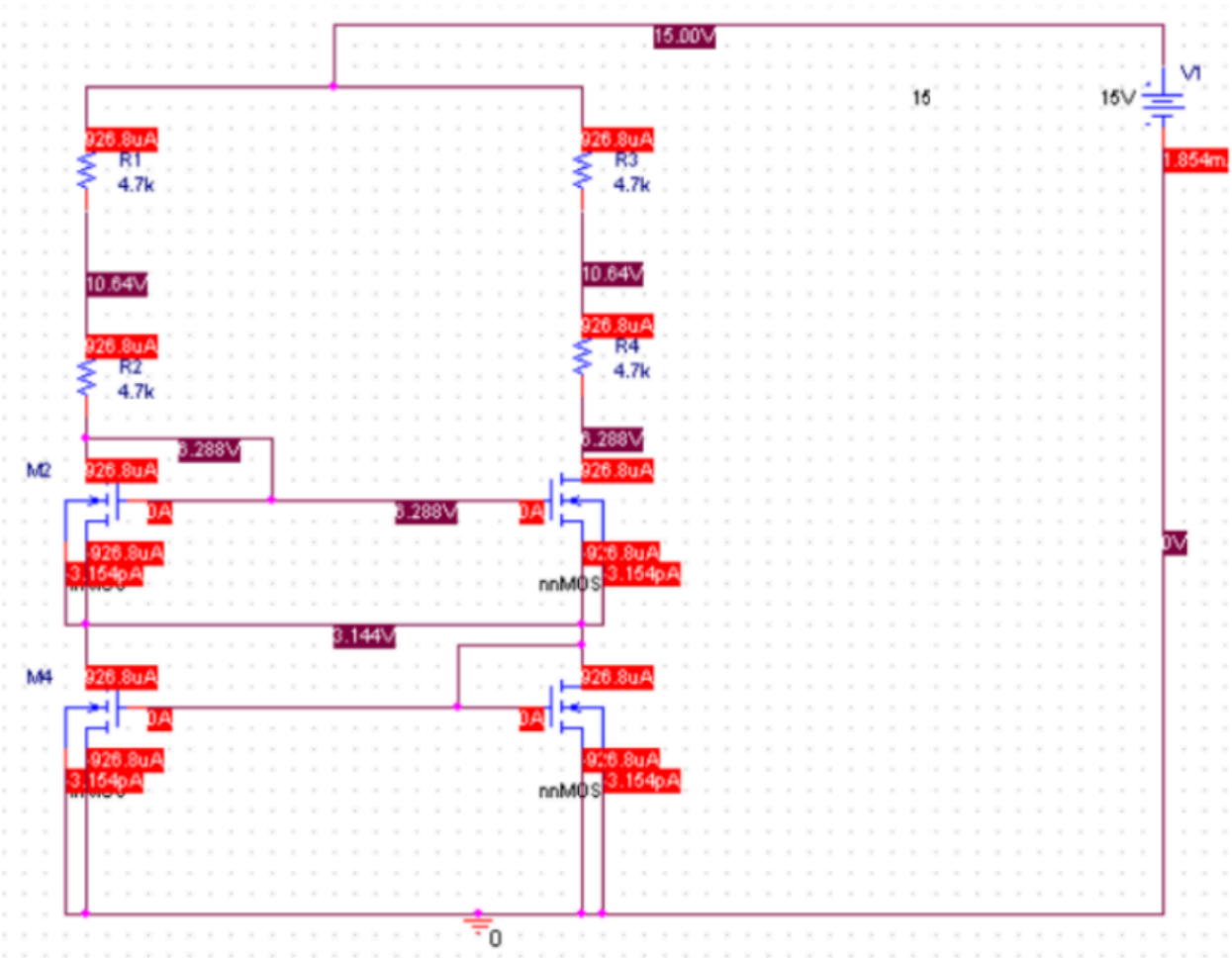
Experiment 11.2




15.00V
V1
15V
1.854mA
926.8uA
R1
4.7k
926.8uA
R3
4.7k
10.64V
10.64V
926.8uA
R2
4.7k
926.8uA
R4
4.7k
6.288V
926.8uA
926.8uA
6.288V
926.8uA
3.154pA
nnMOS
926.8uA
3.154pA
3.144V
926.8uA
926.8uA
926.8uA
3.154pA
nnMOS
926.8uA
3.154pA
0

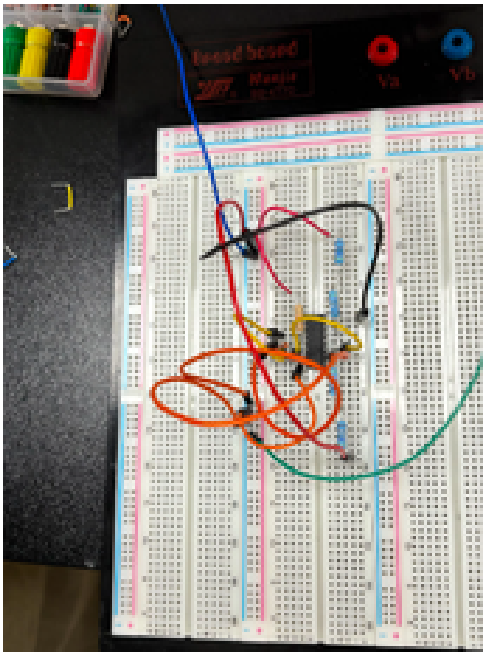


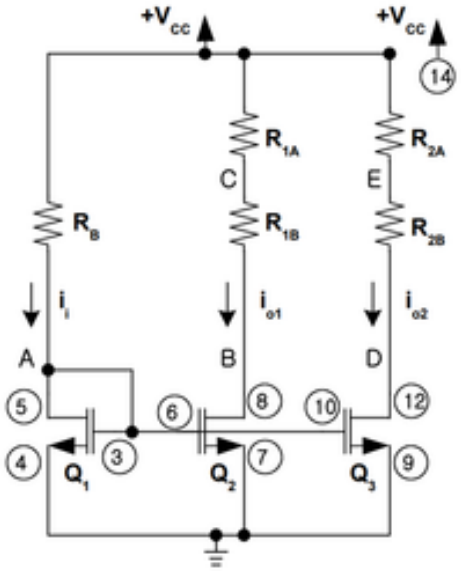
Experiment 11.3:


15.00V
15
15V
V1
926.8uA
R1
4.7k
926.8uA
R3
4.7k
10.64V
10.64V
926.8uA
R2
4.7k
926.8uA
R4
4.7k
M2
926.8uA
926.8uA
926.8uA
3.154pA
nnMOS
926.8uA
3.154pA
M4
926.8uA
926.8uA
926.8uA
3.154pA
nnMOS
926.8uA
3.154pA
0

Main Experiment:

11.1 Create circuit as seen below:



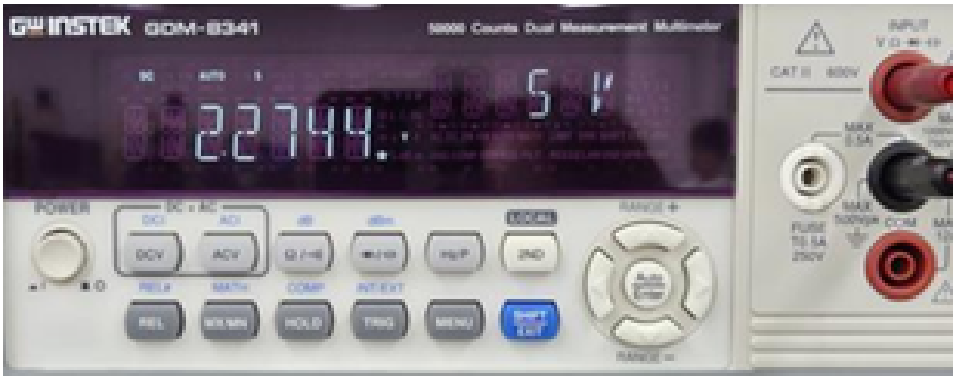


Then measure the terminal voltage and current from A to F by using the multimeter.

Node	A	B	C	D	E
Voltage	2.266	1.532	7.729	4.268	8.091

i1 (mA)	Io1 (mA)	Io2 (mA)	Io1/ii (mA)	Io2/ii (mA)
1.5031	3.0554	3.0638	2.0327	2.0383





Short the R2B and see the voltage at node D to current io2

	Node D voltage	Io2 (mA)	Q3Rout (kOhm)
R2B= 4.7k Ohm	4.2680	3.0638	0.276
R2B= short	4.830	1.0288	

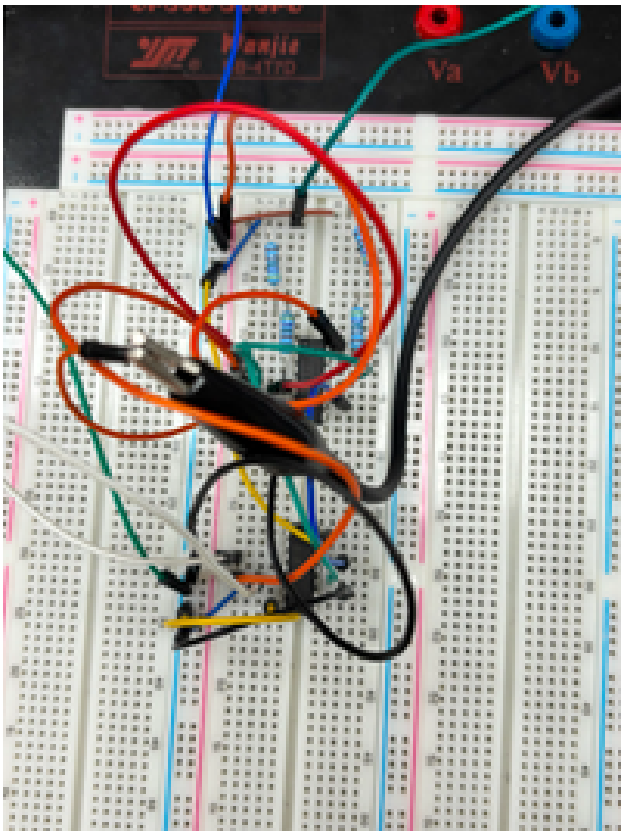
	Node B voltage	Io1 (mA)	Q2Rout (kOhm)
R2B= 4.7k Ohm	1.5322	3.0554	1.423
R2B= short	4.730	0.8083	

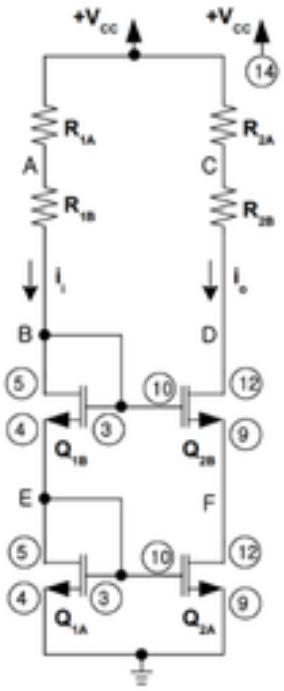




Experiment 11.2: Cascade Current Mirror

1. We built the circuit as shown in the figure.





1. We then measure the terminal voltage and current at A to F using multimeter.

Node	A	B	C	D	E
Voltage	11.75	3.22	12.20	2.76	3.08

i1 (mA)	Io (mA)	Io1/ii (mA)
2.28	2.21	0.969

	Node D voltage	Io (mA)	Rout (kOhm)
R2B= 4.7k Ohm	2.76	2.21	0.966
R2B= short	4.24	2.78	





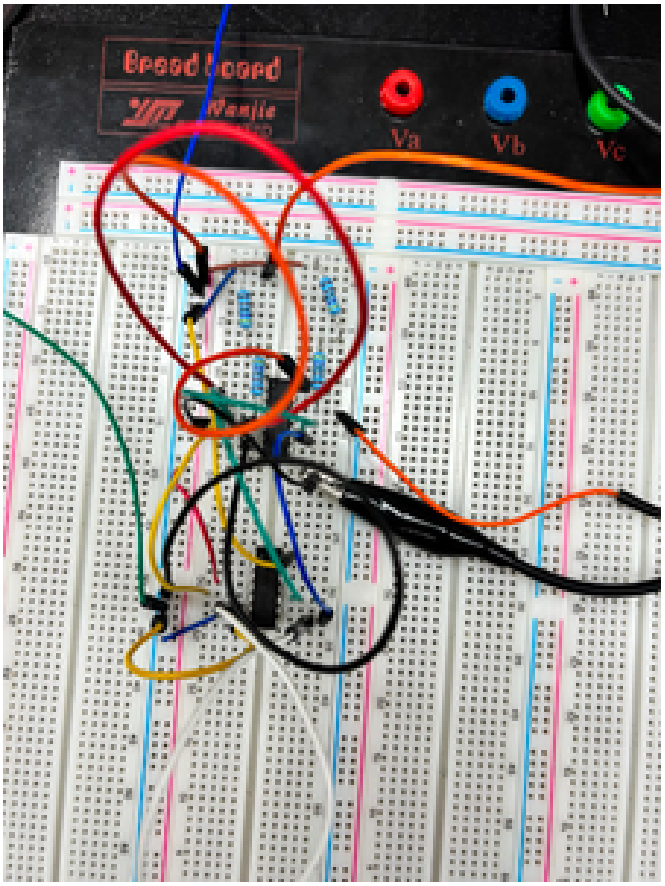

32228.
5 V


30864.
5 V

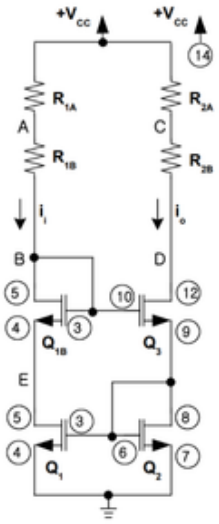

22892.
5mA


22112.
5mA

11.3 Wilson Current Mirror.

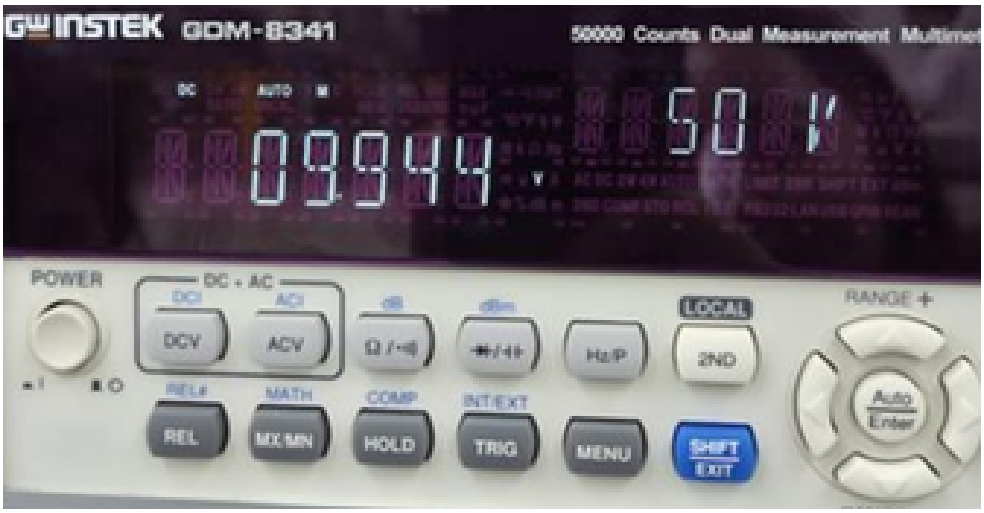


1. The same experiment procedure is then repeated



Measure the terminal voltage and current at A to E using the multimeter

Node	A	B	C	D	E
Voltage	8.693	2.340	8.47	2.36	2.160



i1 (mA)	Io (mA)	Io1/ii (mA)
3.016	3.0325	1.0053

1. We then need to short circuit R2B to measure the voltage at node D and current Io

	Node D voltage	Io (mA)	Rout (kOhm)
R2B= 4.7k Ohm	3.30	2.15	4.582
R2B= short	4.48	2.83	



